spacecraft speed of roughly 1/10 the speed of light—and somewhat higher if we are clever.

In 1968 Freeman Dyson, a brilliant physicist for whom I have great respect, described and analyzed a crude propulsion system that, in the hands of a sufficiently advanced civilization, could achieve this.

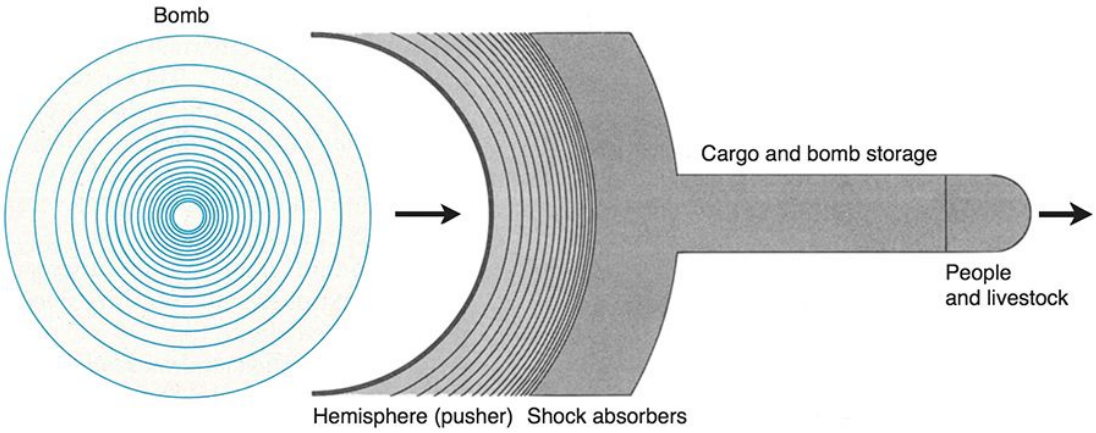


Fig. 13.2. Freeman Dyson's bomb-powered propulsion system. *[From Dyson (1968).]*

Thermonuclear bombs ("hydrogen bombs") are detonated just behind a hemispherical shock absorber that is 20 kilometers in diameter (Figure 13.2). The bomb debris pushes the ship forward, achieving, in Dyson's most optimistic estimate, a speed one-thirtieth that of light. A less crude design could do somewhat better. In 1968 Dyson estimated that such a propulsion system would not be practical any sooner than the late twenty-second century, 150 years from now. I think that's overly optimistic.

Laser Beam and Light Sail

In 1962 Robert Forward, another physicist whom I respect, wrote a short article in a popular magazine about a spacecraft with a sail, pushed